

Contractor's Root Cause Analysis — NCR-NRB4-001

Nandira River Bridge Project — Package NRB-4

Document Reference:	RCA-NRB4-003
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Contract No.:	NHIA/NRB4/CW/2020-01
From:	Contractor, Quality Manager, Sagara Constructions Pvt. Ltd.
To:	Engineer, Meridian Engineering Consultants

Subject: Root Cause Analysis — NCR-NRB4-001, Pier P5 Pier Shaft

In response to NCR-NRB4-001, we have investigated pour records, batching records, curing records, and site photographs for the Pier P5 pier shaft pour of 14-Feb-2022, and identify the following contributing factors.

- CUBE CURING: The curing tank heater fault logged by the Engineer between approximately 18-Feb-2022 and 04-Mar-2022 overlaps the first 9–14 days of curing for the 28-day cubes of this pour, including Test Set B. Reduced curing temperature during this early period is capable of measurably suppressing cube strength gain, particularly for specimens cast later in a working day when ambient temperatures were already falling, which applies to Test Set B (cast 11:40–13:10, the warmest part of the day, with curing tank conditions affected shortly after casting).
- CUBE COMPACTION: Review of site photographs taken during cube-making for this pour indicates the Set B cubes were compacted by a relief technician standing in for the usual laboratory technician, who was on approved leave that day. Compaction of these specific cubes appears less thorough than the standard practice observed for Sets A and C, consistent with minor entrapped air voids visible in the photographs of the Set B specimens after de-moulding.
- IN-PLACE CONCRETE PLACEMENT: We have reviewed the pour record and placement method statement for this pour. The affected portion of the pour was placed and compacted by the standard placing crew using immersion vibrators in accordance with the approved method statement, in the same manner as the remainder of the pour. We have no record or observation indicating that placement or compaction of the in-place concrete itself differed for this portion of the pour.
- CONCLUSION: We consider it likely that the low Test Set B results reflect deficiencies in the cube specimens themselves — curing and compaction — rather than a deficiency in the in-place concrete. We propose additional testing of the in-place concrete (core testing and non-destructive testing) to verify this conclusion, as instructed under NCR-NRB4-001.

Submitted by: Quality Manager, Sagara Constructions Pvt. Ltd.